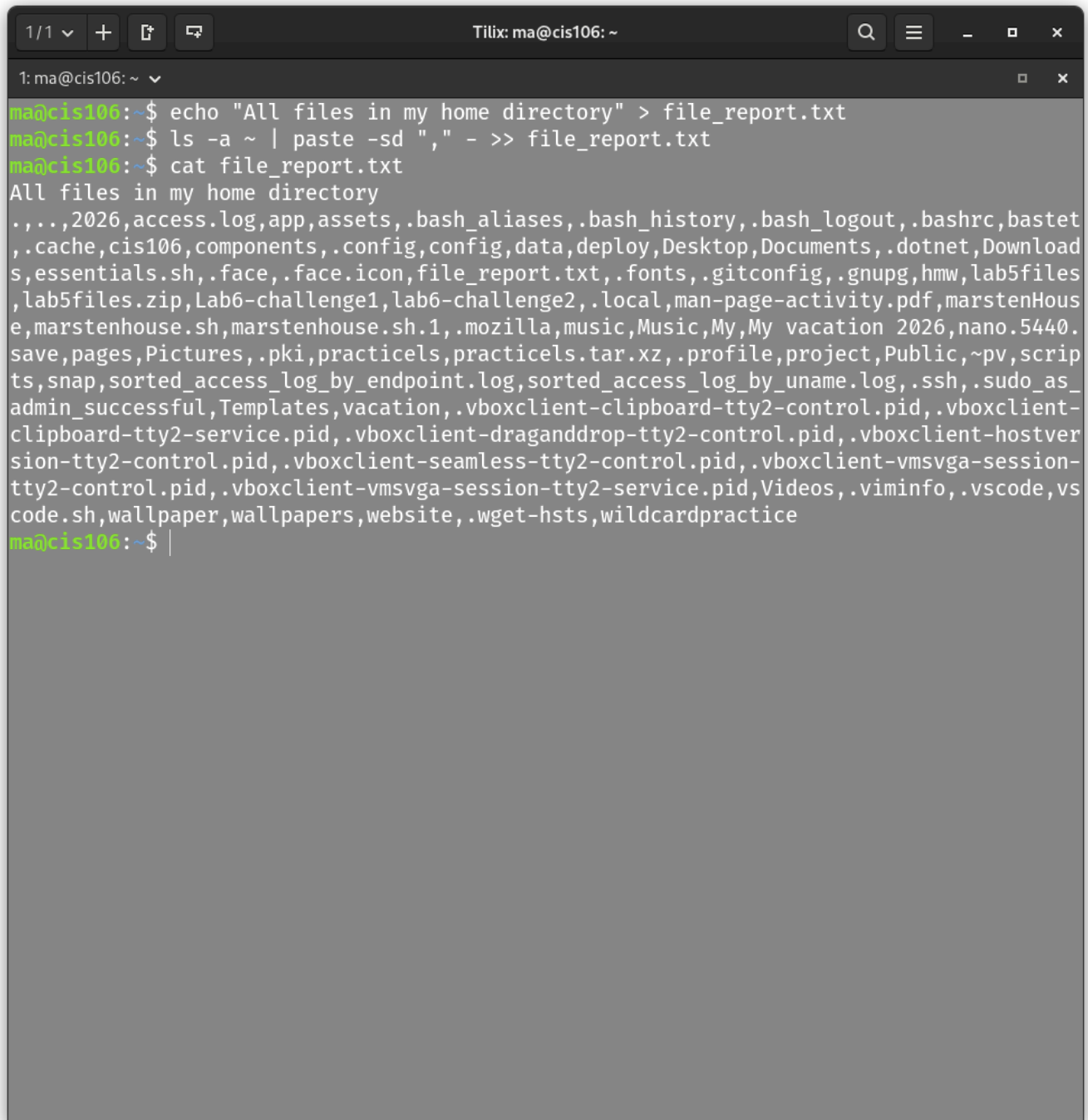


LAB 9

Question 1



```
1/1 + [?] [x] Tilix: ma@cis106: ~
1: ma@cis106: ~
ma@cis106:~$ echo "All files in my home directory" > file_report.txt
ma@cis106:~$ ls -a ~ | paste -sd "," - >> file_report.txt
ma@cis106:~$ cat file_report.txt
All files in my home directory
.,.,.,2026,access.log,app,assets,.bash_aliases,.bash_history,.bash_logout,.bashrc,bastet
,.cache,cis106,components,.config,config,data,deploy,Desktop,Documents,.dotnet,Download
s,essentials.sh,.face,.face.icon,file_report.txt,.fonts,.gitconfig,.gnupg,hmw,lab5files
,lab5files.zip,Lab6-challenge1,lab6-challenge2,.local,man-page-activity.pdf,marstenHous
e,marstenhouse.sh,marstenhouse.sh.1,.mozilla,music,Music,My,My vacation 2026,nano.5440.
save,pages,Pictures,.pki,practicels,practicels.tar.xz,.profile,project,Public,~pv,scrip
ts,snap,sorted_access_log_by_endpoint.log,sorted_access_log_by_uname.log,.ssh,.sudo_as_
admin_successful,Templates,vacation,.vboxclient-clipboard-tty2-control.pid,.vboxclient-
clipboard-tty2-service.pid,.vboxclient-draganddrop-tty2-control.pid,.vboxclient-hostver
sion-tty2-control.pid,.vboxclient-seamless-tty2-control.pid,.vboxclient-vmsvga-session-
tty2-control.pid,.vboxclient-vmsvga-session-tty2-service.pid,Videos,.viminfo,.vscode,vs
code.sh,wallpaper,wallpapers,website,.wget-hsts,wildcardpractice
ma@cis106:~$ |
```

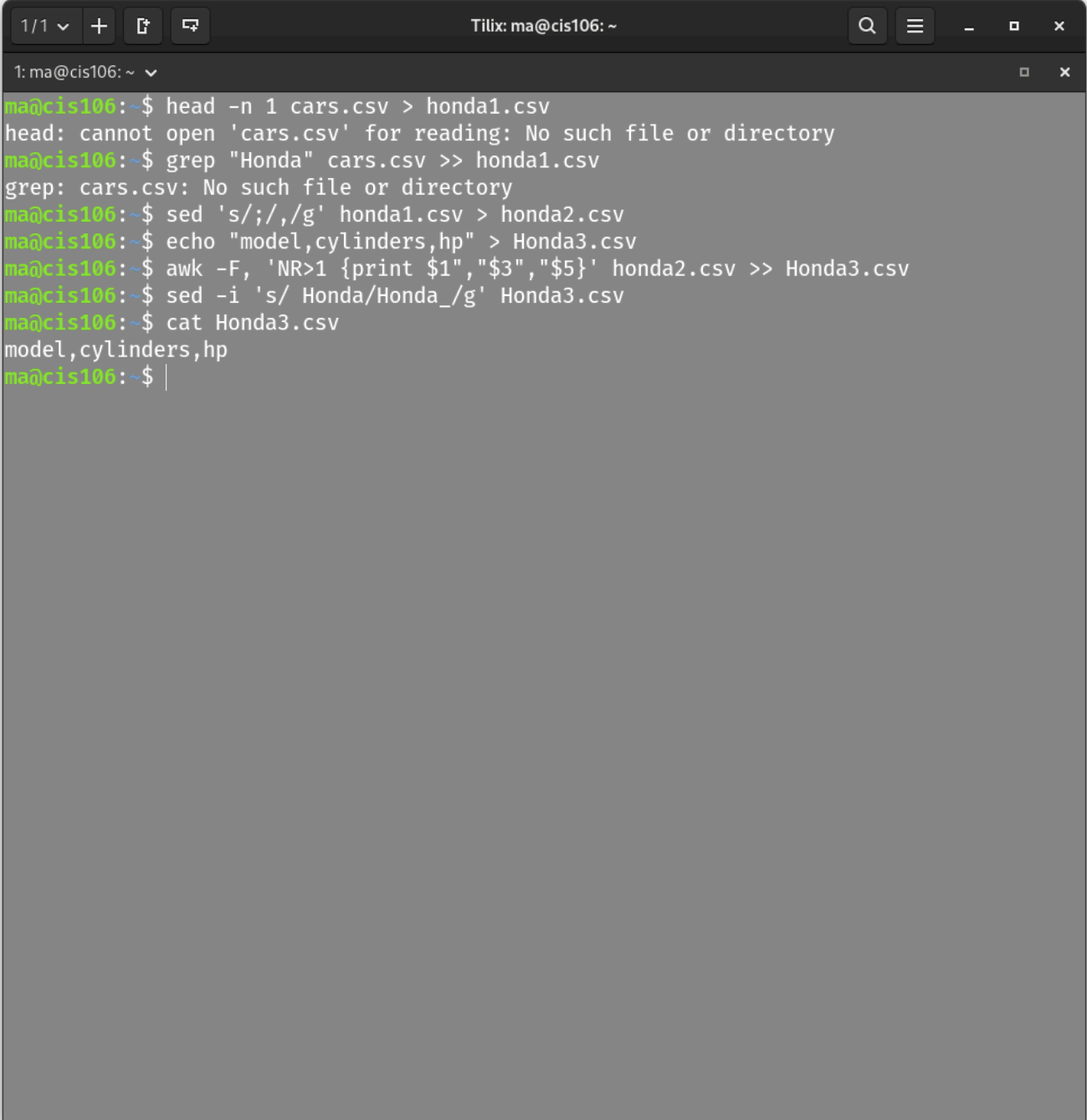
Question 2

```
1/1 + [ ] [ ]
Tilix: ma@cis106: ~
1: ma@cis106: ~
ma@cis106:~$ ls -a ~ | paste -sd "," - >> file_report.txt
ma@cis106:~$ cat file_report.txt
All files in my home directory
.,.,.,2026,access.log,app,assets,.bash_aliases,.bash_history,.bash_logout,.bashrc,bastet
,.cache,cis106,components,.config,config,data,deploy,Desktop,Documents,.dotnet,Download
s,essentials.sh,.face,.face.icon,file_report.txt,.fonts,.gitconfig,.gnupg,hmw,lab5files
,lab5files.zip,Lab6-challenge1,lab6-challenge2,.local,man-page-activity.pdf,marstenHous
e,marstenhouse.sh,marstenhouse.sh.1,.mozilla,music,Music,My,My vacation 2026,nano.5440.
save,pages,Pictures,.pki,practicels,practicels.tar.xz,.profile,project,Public,~pv,scrip
ts,snap,sorted_access_log_by_endpoint.log,sorted_access_log_by_uname.log,.ssh,.sudo_as_
admin_successful,Templates,vacation,.vboxclient-clipboard-tty2-control.pid,.vboxclient-
clipboard-tty2-service.pid,.vboxclient-draganddrop-tty2-control.pid,.vboxclient-hostver
sion-tty2-control.pid,.vboxclient-seamless-tty2-control.pid,.vboxclient-vmvga-session-
tty2-control.pid,.vboxclient-vmvga-session-tty2-service.pid,Videos,.viminfo,.vscode,vs
code.sh,wallpaper,wallpapers,website,.wget-hsts,wildcardpractice
ma@cis106:~$ echo "Total number of files in my home directory" >> file_report.txt
ma@cis106:~$ ls -lA ~ | grep "^-" | wc -l >> file_report.txt
ma@cis106:~$ echo "Total number of directories in my home directory" >> file_report.txt
ma@cis106:~$ ls -lA ~ | grep "^d" | wc -l >> file_report.txt
ma@cis106:~$ tail -n 10 file_report.txt
All files in my home directory
.,.,.,2026,access.log,app,assets,.bash_aliases,.bash_history,.bash_logout,.bashrc,bastet
,.cache,cis106,components,.config,config,data,deploy,Desktop,Documents,.dotnet,Download
s,essentials.sh,.face,.face.icon,file_report.txt,.fonts,.gitconfig,.gnupg,hmw,lab5files
,lab5files.zip,Lab6-challenge1,lab6-challenge2,.local,man-page-activity.pdf,marstenHous
e,marstenhouse.sh,marstenhouse.sh.1,.mozilla,music,Music,My,My vacation 2026,nano.5440.
save,pages,Pictures,.pki,practicels,practicels.tar.xz,.profile,project,Public,~pv,scrip
ts,snap,sorted_access_log_by_endpoint.log,sorted_access_log_by_uname.log,.ssh,.sudo_as_
admin_successful,Templates,vacation,.vboxclient-clipboard-tty2-control.pid,.vboxclient-
clipboard-tty2-service.pid,.vboxclient-draganddrop-tty2-control.pid,.vboxclient-hostver
sion-tty2-control.pid,.vboxclient-seamless-tty2-control.pid,.vboxclient-vmvga-session-
tty2-control.pid,.vboxclient-vmvga-session-tty2-service.pid,Videos,.viminfo,.vscode,vs
code.sh,wallpaper,wallpapers,website,.wget-hsts,wildcardpractice
Total number of files in my home directory
30
Total number of directories in my home directory
45
ma@cis106:~$ |
```

Question 3

```
Tilix: ma@cis106: ~
1: ma@cis106: ~
The user: uucp logs in with: /usr/sbin/nologin and is home dir is: /var/spool/uucp
The user: proxy logs in with: /usr/sbin/nologin and is home dir is: /bin
The user: www-data logs in with: /usr/sbin/nologin and is home dir is: /var/www
The user: backup logs in with: /usr/sbin/nologin and is home dir is: /var/backups
The user: list logs in with: /usr/sbin/nologin and is home dir is: /var/list
The user: irc logs in with: /usr/sbin/nologin and is home dir is: /run/ircd
The user: _apt logs in with: /usr/sbin/nologin and is home dir is: /nonexistent
The user: nobody logs in with: /usr/sbin/nologin and is home dir is: /nonexistent
The user: systemd-network logs in with: /usr/sbin/nologin and is home dir is: /
The user: dhcpcd logs in with: /bin/false and is home dir is: /usr/lib/dhcpcd
The user: tss logs in with: /bin/false and is home dir is: /var/lib/tpm
The user: systemd-timesync logs in with: /usr/sbin/nologin and is home dir is: /
The user: messagebus logs in with: /usr/sbin/nologin and is home dir is: /nonexistent
The user: dnsmasq logs in with: /usr/sbin/nologin and is home dir is: /var/lib/misc
The user: avahi logs in with: /usr/sbin/nologin and is home dir is: /run/avahi-daemon
The user: speech-dispatcher logs in with: /bin/false and is home dir is: /run/speech-dispatcher
The user: usbmux logs in with: /usr/sbin/nologin and is home dir is: /var/lib/usbmux
The user: cups-pk-helper logs in with: /usr/sbin/nologin and is home dir is: /nonexistent
The user: fwupd-refresh logs in with: /usr/sbin/nologin and is home dir is: /var/lib/fwupd
The user: geoclue logs in with: /usr/sbin/nologin and is home dir is: /var/lib/geoclue
The user: gnome-remote-desktop logs in with: /usr/sbin/nologin and is home dir is: /var/lib/gnome-remote-desktop
The user: saned logs in with: /usr/sbin/nologin and is home dir is: /var/lib/saned
The user: polkitd logs in with: /usr/sbin/nologin and is home dir is: /
The user: rtkit logs in with: /usr/sbin/nologin and is home dir is: /proc
The user: colord logs in with: /usr/sbin/nologin and is home dir is: /var/lib/colord
The user: Debian-gdm logs in with: /bin/false and is home dir is: /var/lib/gdm3
The user: ma logs in with: /bin/bash and is home dir is: /home/ma
The user: _flatpak logs in with: /usr/sbin/nologin and is home dir is: /nonexistent
The user: vboxadd logs in with: /bin/false and is home dir is: /var/run/vboxadd
ma@cis106:~$ awk -F: '{print "The user: " $1 " logs in with: " $7 " and is home dir is: " $6}' /etc/passwd > report_users.txt
ma@cis106:~$ grep "/bin/bash" /etc/passwd | awk -F: '{print $1 " = " $7}' >> report_users.txt
ma@cis106:~$ |
```

Question 4



```
1/1 v + [ ] [ ] Tilix: ma@cis106: ~
1: ma@cis106: ~ v
ma@cis106:~$ head -n 1 cars.csv > honda1.csv
head: cannot open 'cars.csv' for reading: No such file or directory
ma@cis106:~$ grep "Honda" cars.csv >> honda1.csv
grep: cars.csv: No such file or directory
ma@cis106:~$ sed 's/;/,/g' honda1.csv > honda2.csv
ma@cis106:~$ echo "model,cylinders,hp" > Honda3.csv
ma@cis106:~$ awk -F, 'NR>1 {print $1,"$3","$5}' honda2.csv >> Honda3.csv
ma@cis106:~$ sed -i 's/ Honda/Honda_/g' Honda3.csv
ma@cis106:~$ cat Honda3.csv
model,cylinders,hp
ma@cis106:~$ |
```

Question 5

The screenshot shows a terminal window with two panes. The left pane displays the contents of the file `/home/ma/.bash_aliases`. The right pane shows the execution of several alias commands in a shell.

Left Pane: /home/ma/.bash_aliases

```
#-----#
#- Package Management -#
#-----#
alias update='sudo apt update; sudo apt up>'
alias search='apt search '
alias install='sudo apt install -y '
alias sininstall='sudo snap install '
alias pininstall='pip3 install '
alias remove='sudo apt autoremove -y'
alias clean='sudo apt autoremove -y; sudo >'
alias totalpack='dpkg --get-selections | w>'
alias isinstalled='dpkg --get-selections | >'
#-----#
#---- Ls Aliases & defaults ----#
#-----#
if [ -x /usr/bin/dircolors ]; then
    test -r ~/.dircolors && eval "$(dircol>
    alias ls='ls --color=auto'
    alias dir='dir --color=auto'
    alias vdir='vdir --color=auto'
    alias grep='grep --color=auto'
    alias fgrep='fgrep --color=auto'
    alias egrep='egrep --color=auto'
fi
alias ll='ls -alF'
alias la='ls -a'
alias lA='ls -A'
alias lr='ls -R'
alias lsoptions="man ls | grep '^[:space:>'
alias lshelp="ls --help | grep '^[:space:>'
alias lsnodir="ls -FSsh | grep -v '/'"
alias lsdirsize="du -h --max-depth=1"
alias lsnoperm="ls -GAhtml --color=always>'
#-----#
```

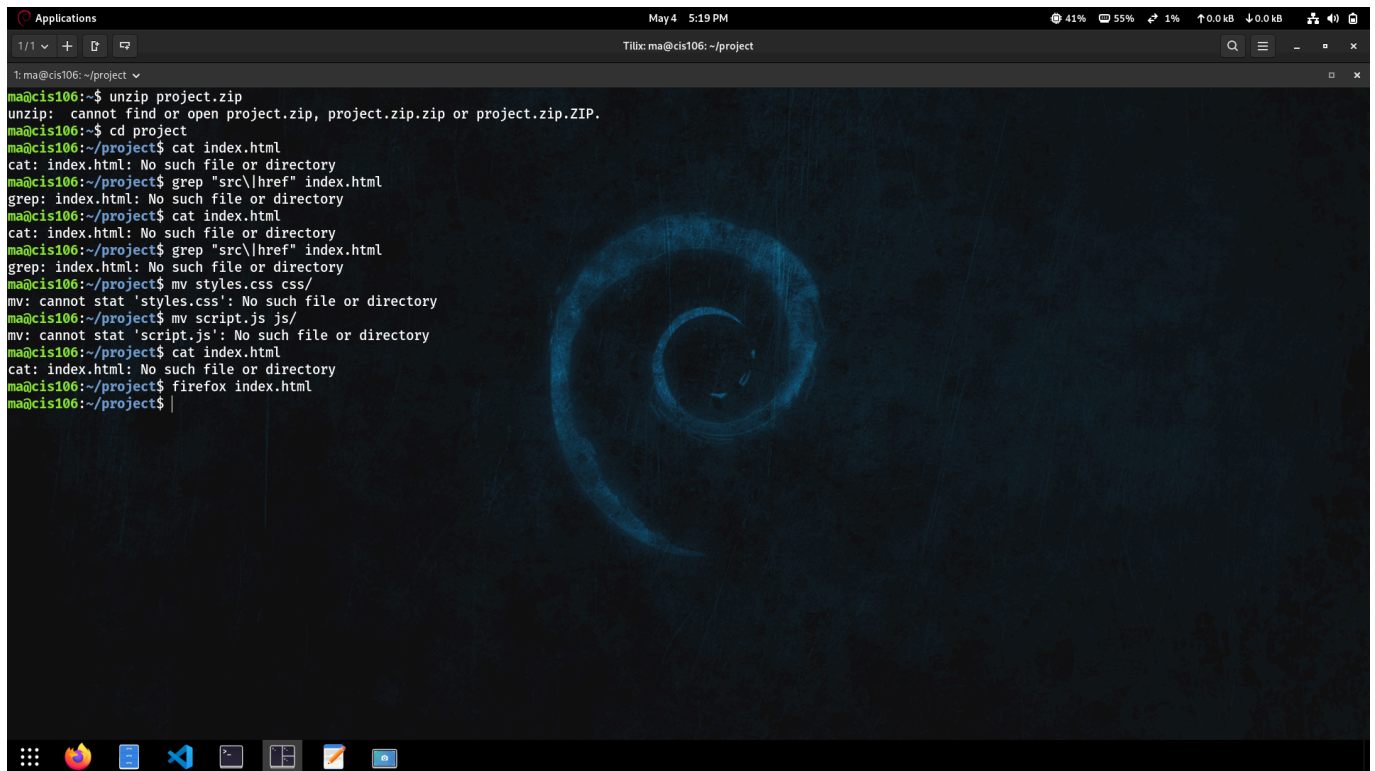
Right Pane: Shell Execution

```
ma@cis106:~$ alias gotocis="cd $HOME/cis106"
ma@cis106:~$ alias update="sudo apt update"
ma@cis106:~$ alias upgrade="sudo apt upgrad
e"
ma@cis106:~$ alias install="sudo apt instal
l"
ma@cis106:~$ alias clean="sudo apt autoremo
ve -y; sudo apt autoclean; sudo apt purge"
ma@cis106:~$ alias lsx="exa -T -l --sort=ex
t --no-permissions --no-user --no-time"
ma@cis106:~$ source ~/.bashrc
ma@cis106:~$
```

Terminal Footer:

```
^G Help    ^O Write O ^F Where I ^K Cut
^X Exit    ^R Read Fi ^\ Replace ^U Paste
```

challenge Question



```
Applications May 4 5:19 PM 41% 55% 1% 0.0 KB 0.0 KB
Tilix: ma@cis106: ~/project
ma@cis106:~/project$ unzip project.zip
unzip: cannot find or open project.zip, project.zip.zip or project.zip.ZIP.
ma@cis106:~/project$ cd project
ma@cis106:~/project$ cat index.html
cat: index.html: No such file or directory
ma@cis106:~/project$ grep "src\\href" index.html
grep: index.html: No such file or directory
ma@cis106:~/project$ cat index.html
cat: index.html: No such file or directory
ma@cis106:~/project$ grep "src\\href" index.html
grep: index.html: No such file or directory
ma@cis106:~/project$ mv styles.css css/
mv: cannot stat 'styles.css': No such file or directory
ma@cis106:~/project$ mv script.js js/
mv: cannot stat 'script.js': No such file or directory
ma@cis106:~/project$ cat index.html
cat: index.html: No such file or directory
ma@cis106:~/project$ firefox index.html
ma@cis106:~/project$
```